

```
ankur@ankur-laptop:~/Documents/Training/geekzine & lfy/lfy/mysqlypyphp$ python mysql.py
Displaying All the Rows : (('Foss Bytes', 'LFY Team'), ('Connecting MySQL', 'Ankur Aggarwal'))
('Foss Bytes', 'LFY Team')
Displaying the first row : ('Foss Bytes', 'LFY Team')
No of rows : 2
ankur@ankur-laptop:~/Documents/Training/geekzine & lfy/lfy/mysqlypyphp$
```

Figure 1: Output of the Python script

```
Array
(
    [0] => Foss Bytes
    [name] => Foss Bytes
    [1] => LFY Team
    [author] => LFY Team
)
Array
(
    [0] => Connection MySQL
    [name] => Connection MySQL
    [1] => Ankur Aggarwal
    [author] => Ankur Aggarwal
)
```

Figure 2: Output of the PHP script

2. We connected to the MySQL server by calling the *MySQLdb connect()* function. The parameters required were the server name, MySQL username, MySQL password, and the database you want to use.
3. To retrieve query results, we created *cur*, an object of the *cursor()* class, which will be responsible for execution and fetching.
4. *cur.execute()* executes the query to create a table named “lfy”.
5. We inserted values into the *lfy* table.
6. We retrieved all values in the *lfy* table through a *SELECT* query.
7. *cur.fetchall()* will fetch all results of the query, and return them as a nested list.
8. *cur.fetchone()* will fetch one row at a time. You might wonder why we executed the *SELECT* query again—because *fetchall()* has already fetched all results of the previous query, and calling *fetchone()* would return *None*.
9. The *rowcount* property tells us the number of rows returned by the query.
10. We closed the cursor object, freeing the resources it holds.
11. We closed the connection. Always remember to close the connections; otherwise, it may be a major security risk.

 **Note:** Python being an interpreted language, it executes code line by line. If an error occurred on the 8th line, by then, it has executed the first seven lines, creating the table and inserting entries. The connection also remains open. Therefore, I recommend you adopt error and exception handling while using databases.

Connecting MySQL with PHP

PHP is very popular for server-side scripting, and MySQL databases are widely used for storage of the data used in dynamic pages. Connecting to MySQL from PHP is quite easy, with in-built functions; it follows the same four rules we discussed in the Python section. We will do the same basic

tasks as the Python script; so before trying out the code below, drop the previous table. Here's our PHP script, *mysqlphp.php*:

```
<?php
$conn=mysql_connect("servername","username","password") or
die("connectivity failed"); #1
mysql_select_db("database name",$conn); #2
$sql="create table lfy(name varchar(50), author varchar(50))";
mysql_query($sql,$conn); #3
$insert="insert into lfy values('Foss Bytes','LFY Team')";
$insert2="insert into lfy values('Connection MySQL','Ankur
Aggarwal')";
mysql_query($insert,$conn);
mysql_query($insert2,$conn);
$sql="select * from lfy";
while($row=mysql_fetch_array($sql)) #4
{
    print_r($row);
}
#5
mysql_close($conn); #6
?>
```

A detailed explanation of the above script is given below:

1. If the connection to MySQL fails, we use the *die()* function as an error-handling technique, to stop processing the page. The connection object is passed to other MySQL-related functions for use.
2. We selected the database we want to use.
3. We used *mysql_query()* to execute queries.
4. *mysql_fetch_array()* will fetch the results of the query in the form of an array. We have used a *while* loop to display the results.
5. To display the data, we used *print_r()*, which will display the whole row at one go. To access individual fields, use a string index of the field-name, like *row["name"]* etc. See the output screenshot (Figure 2) for a clearer understanding.
6. We closed the connection to the MySQL server. Connecting to a database is easy! So start playing with your database through scripting. :D 

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The author loves to explore open source technologies, and enjoys programming. He has learned C, C++, core Java, and Python. He loves rock/heavy metal music, and has been trying his hand at the guitar. He has also started a blog recently (www.fossstuff.wordpress.com) and can be contacted via coolankur2006@gmail.com, or on Twitter as @ankurtwi.